Special Participation E — AI-Enhanced Learning Tool

Title: µP & Modern Optimizers Coach (EECS 182/282)
Link: µP & Modern Optimizers Coach

1. Purpose

This project explored whether a Custom GPT could serve as an active learning companion for deep-learning optimization topics, especially µP (Maximal Update Parameterization), RMSNorm, and optimizers such as AdamW, Lion, and Muon.

Instead of static pre-lecture readings, the goal was to test if structured, conversational explanations could build intuition for how:

- RMS, Frobenius, and Spectral norms describe update magnitudes,
- Optimizer dynamics behave as locally linear transformations, and
- µP scaling stabilizes learning across model widths.

2. Creation Process

Began with a standard ChatGPT chat to probe how well a generic model could explain RMS/Frobenius/Spectral norms and optimizer dynamics.

→ **Trace 1:** I found that explanations quickly became verbose and inconsistent—messy for me to keep track of. I decided to build a Custom GPT using the "Explore → Create GPT" flow on `chatgpt.com`.

Knowledge uploads

- Selected pages from *Understanding Deep Learning* (Prince, 2025)
- Relevant homework and discussion solutions
- A PDF of my earlier ChatGPT conversation on RMS / Frobenius / Spectral norms

(Lecture slides were skipped because text couldn't be extracted.)

Enabled code and reasoning tools to allow quantitative explanations.

Configured it with a structured prompt style:

- Explain → Test → Reflect teaching loop
- Concise first answer with expandable "details" sections
- Quiz and comparison modes for self-testing

3. Evaluation and Interaction Trace

Stage	Example Prompt	Observation
1. Concept Understanding	"Give me a 10-minute crash course on P with examples."	Clear and well-organized overview of scaling laws and update-norm invariance. Used formulas from Prince and Yang et al. (2022).
2. Application / Example	"Design a P-aware LR sweep for CIFAR-10 using AdamW and Lion."	Produced a realistic grid and justified parameter scaling. Highlighted Lion's sign-based efficiency.
3. Critical Questioning	"Explain where P fails or why it might diverge."	Gave nuanced discussion of regime limits and mis-scaled initialization. No hallucinations detected.

Traces

Session 1
Session 2
Session 3

4. Critical Evaluation

I deliberately tested borderline misconceptions (e.g., "P makes training fully invariant to width."). The GPT correctly qualified its answer, noting that invariance is approximate and dependent on consistent scaling.

Grounding it in course artifacts kept terminology accurate and prevented hallucinations; minor stylistic drift appeared only around SOAP. Even without forcing an error, I showed critical checking by identifying where risk of over-generalization could appear and verifying against the course book.

5. Reflection and Future Use

Creating a domain-specific GPT proved significantly more effective than generic chat interaction. It provided structured, layered explanations aligned with lecture objectives and functioned like a Socratic tutor.

Key insights:

- Context files (Concept Index + Identity Prompt) greatly shape teaching quality.
- Course artifacts anchor vocabulary and reduce hallucination risk.
- Verification over generation still deepens learning.

6. Broader Applicability

The same approach could scale to the entire EECS 182/282 curriculum by creating specialized GPTs for each core theme (e.g., Autoencoders, CNNs, RNNs, Transformers, Interpretability).

Each model could include targeted readings, homework snippets, and quiz banks—effectively replacing static pre-lecture notes with interactive study companions.

Over time, a shared library of course-aligned GPTs could enable students to revisit any concept through guided dialogue and personalized testing, turning AI tools into an integrated learning ecosystem.